

Project Title - Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

Problem Statement: Weather fluctuations, farming practices, and input utilization create inconsistent seasonal agricultural yield and financial performance.

Core Objectives:

- Identify seasonal patterns in environmental conditions, soil health, and resource consumption.
- Evaluate crop yield and economic outcomes (costs, revenues, profits) across cropping seasons.
- Deliver evidence-based recommendations for resource management, risk reduction, and profit maximization.



Project Description:



The Seasonal Agriculture Performance Analysis project is a comprehensive data analytics initiative examining 4,000 farm-level observation records across 8 Indian states and 10 districts. By synthesizing environmental metrics (rainfall, temperature, humidity, sunlight), soil health factors (pH, moisture, NPK), farm input choices (irrigation methods, seed quality, fertilizers, pesticides), and production outputs, the project evaluates crop yield and financial viability across the Kharif, Rabi, and Zaid cropping cycles. The primary focus is to identify key drivers of agricultural profitability, model operational risk (such as disease and pest outbreaks), and optimize resource efficiency—particularly water and nutrient application—to promote sustainable and profitable farming practices.



WHO ARE THE END USERS?

- Farmers & Farm Managers: Benefit from actionable insights on optimal seasonal crop selection, precision input usage, and efficient irrigation techniques (e.g., Drip vs. Flood) to lower input costs and boost net profitability.
- Agricultural Extension Officers: Rely on localized statistical findings regarding soil parameters, pest risks, and resource consumption to deliver targeted, evidence-based recommendations to farming communities.
- AgTech & Agri-Finance Organizations: Utilize profit, revenue, and yield metrics to design tailored agricultural technology tools, perform data-backed credit risk assessments, and structure crop insurance products.
- Policy Makers & Regional Planners: Leverage state and district-level comparative data to guide resource allocation, inform agricultural subsidy programs, and design effective water and soil management initiatives.

Technology Used

- Python
- Matplotlib
- Seaborn
- Pandas
- numpy
- google colab



RESULTS

```
#Load the dataset
df = pd.read_csv("/content/drive/MyDrive/seasonal_agriculture_performance_dataset.csv")
#top five rows
df.head()
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct	Nitrogen_kg_ha	Phosphorus_kg_ha
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.40	24.30	69.10	5.80	7.14	32.80	75.80	61.30
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.40	25.70	78.40	5.10	5.20	32.00	82.10	35.40
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.60	23.90	56.40	10.30	5.82	8.10	118.50	51.50
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.20	29.70	65.20	6.30	5.98	23.70	138.60	55.60
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.60	34.10	55.20	6.50	6.90	29.30	124.10	69.00

```
#Bottom five rows
df.tail(5)
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct	Nitrogen_kg_ha	Phosphorus_kg_ha
3995	SF13996	Karnataka	Rajkot	Wheat	Kharif	4.72	865.40	27.30	51.40	6.30	7.97	27.30	104.30	44.80
3996	SF13997	Karnataka	Krishna	Chilli	Kharif	8.77	947.50	29.20	85.50	6.10	6.57	33.70	127.80	44.90
3997	SF13998	Andhra Pradesh	Erode	Pulses	Zaid	10.47	80.00	35.40	68.50	8.60	7.25	21.10	126.70	59.90
3998	SF13999	Punjab	Warangal	Cotton	Kharif	1.73	475.00	28.20	61.70	6.40	6.11	30.50	107.50	20.70
3999	SF14000	Punjab	Nashik	Wheat	Kharif	3.31	984.30	25.80	77.20	5.90	6.16	35.70	137.10	92.60

RESULTS

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 4000 entries, 0 to 3999
```

```
Data columns (total 28 columns):
```

#	Column	Non-Null Count	Dtype
0	Farm_ID	4000 non-null	object
1	State	4000 non-null	object
2	District	4000 non-null	object
3	Crop	4000 non-null	object
4	Season	4000 non-null	object
5	Farm_Area_Hectares	4000 non-null	float64
6	Rainfall_mm	3952 non-null	float64
7	Avg_Temperature_C	4000 non-null	float64
8	Humidity_pct	4000 non-null	float64
9	Sunlight_Hours_Day	4000 non-null	float64
10	Soil_pH	4000 non-null	float64
11	Soil_Moisture_pct	3960 non-null	float64
12	Nitrogen_kg_ha	4000 non-null	float64
13	Phosphorus_kg_ha	4000 non-null	float64
14	Potassium_kg_ha	4000 non-null	float64
15	Irrigation_Method	4000 non-null	object
16	Fertilizer_kg_ha	4000 non-null	float64
17	Pesticide_Litre_ha	4000 non-null	float64
18	Seed_Quality_Score	4000 non-null	float64
19	Yield_Tonnes_Ha	3968 non-null	float64
20	Production_Tonnes	4000 non-null	float64
21	Market_Price_INR_Tonne	4000 non-null	int64
22	Total_Cost_INR	4000 non-null	int64
23	Revenue_INR	4000 non-null	int64
24	Profit_INR	4000 non-null	int64
25	Water_Used_m3	4000 non-null	int64
26	Water_Efficiency_t_per_1000m3	4000 non-null	float64
27	Disease_Pest_Risk_pct	4000 non-null	float64

```
dtypes: float64(17), int64(5), object(6)
```

RESULTS

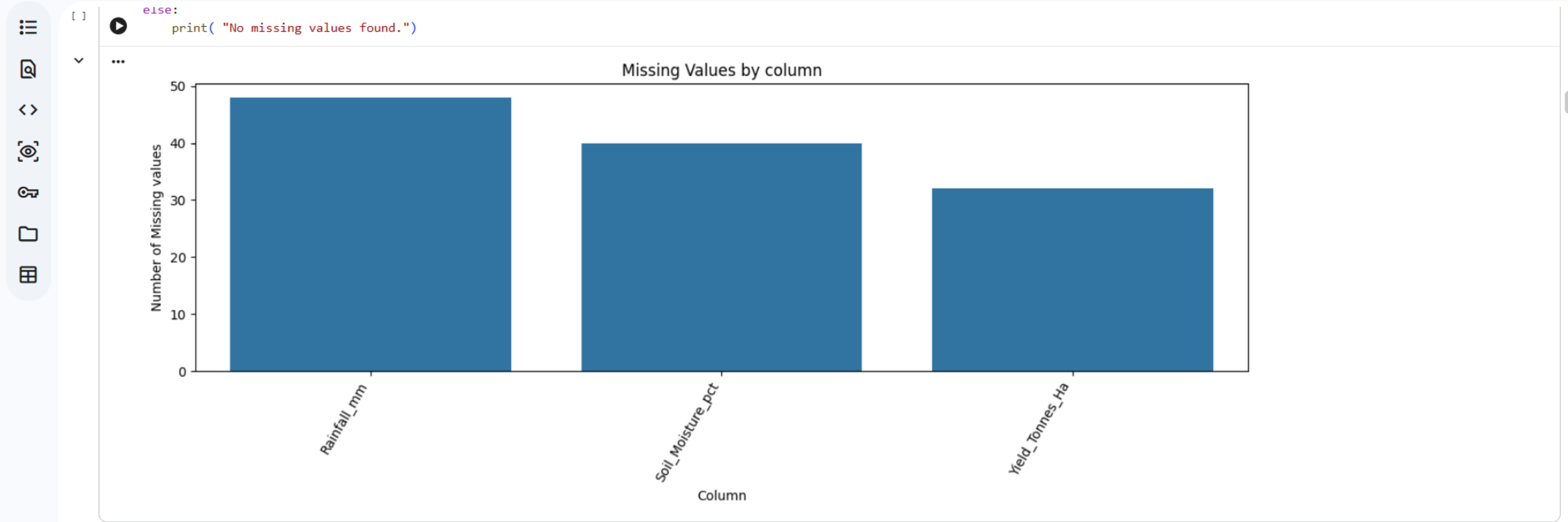
 Major Project: Seasonal Agriculture Performance Analysis .ipynb

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RAM Disk



5. Duplicate Records - Identification and Handling

RFS II TS



Major Project: Seasonal Agriculture Performance Analysis .ipynb



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Reset



RAM Disk



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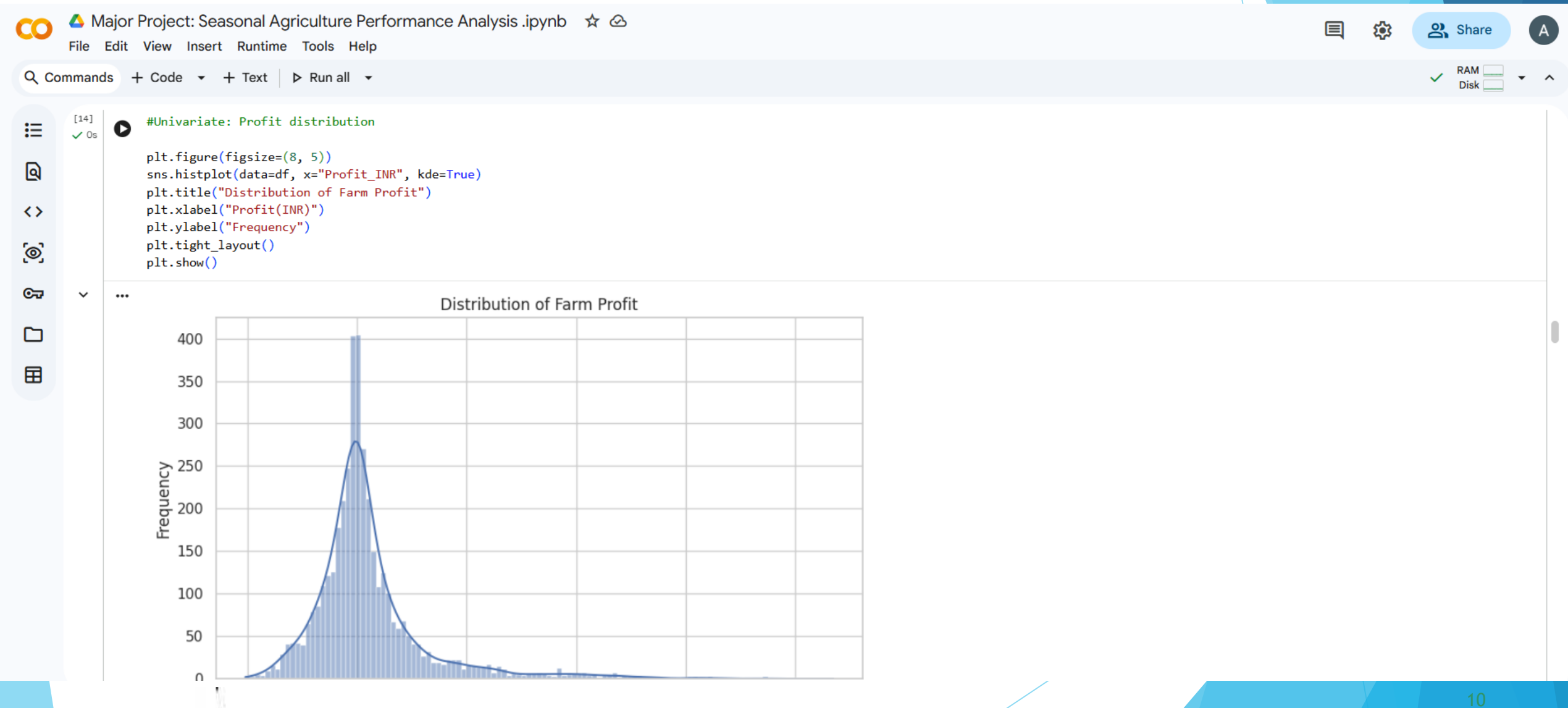
#Seasonal descriptive summary

```
season_summary = df.groupby("Season")[numeric_columns].agg(["mean", "median", "std"])
season_summary
```

	Farm_Area_Hectares			Rainfall_mm			Avg_Temperature_C			Humidity_pct	...	Profit_INR
	mean	median	std	mean	median	std	mean	median	std	mean	...	std
Season												
Kharif	7.944924	8.180	4.282328	852.084205	854.75	182.552740	28.454019	28.5	2.510166	71.812591	...	611549.783883
Rabi	7.752207	7.680	4.179402	436.000124	430.30	175.659664	23.489183	23.5	2.459561	57.893178	...	489901.254255
Zaid	8.091246	8.085	4.162594	299.421784	283.30	153.980457	31.042088	31.0	2.554911	52.012121	...	436356.747261

3 rows × 66 columns

RESULTS



Future scope

1. Predictive Analytics & Machine Learning
2. Real-Time IoT & Smart Farming Integration
3. Geospatial & Remote Sensing Intelligence
4. Financial & Market Intelligence
5. Accessible Decision Support Systems

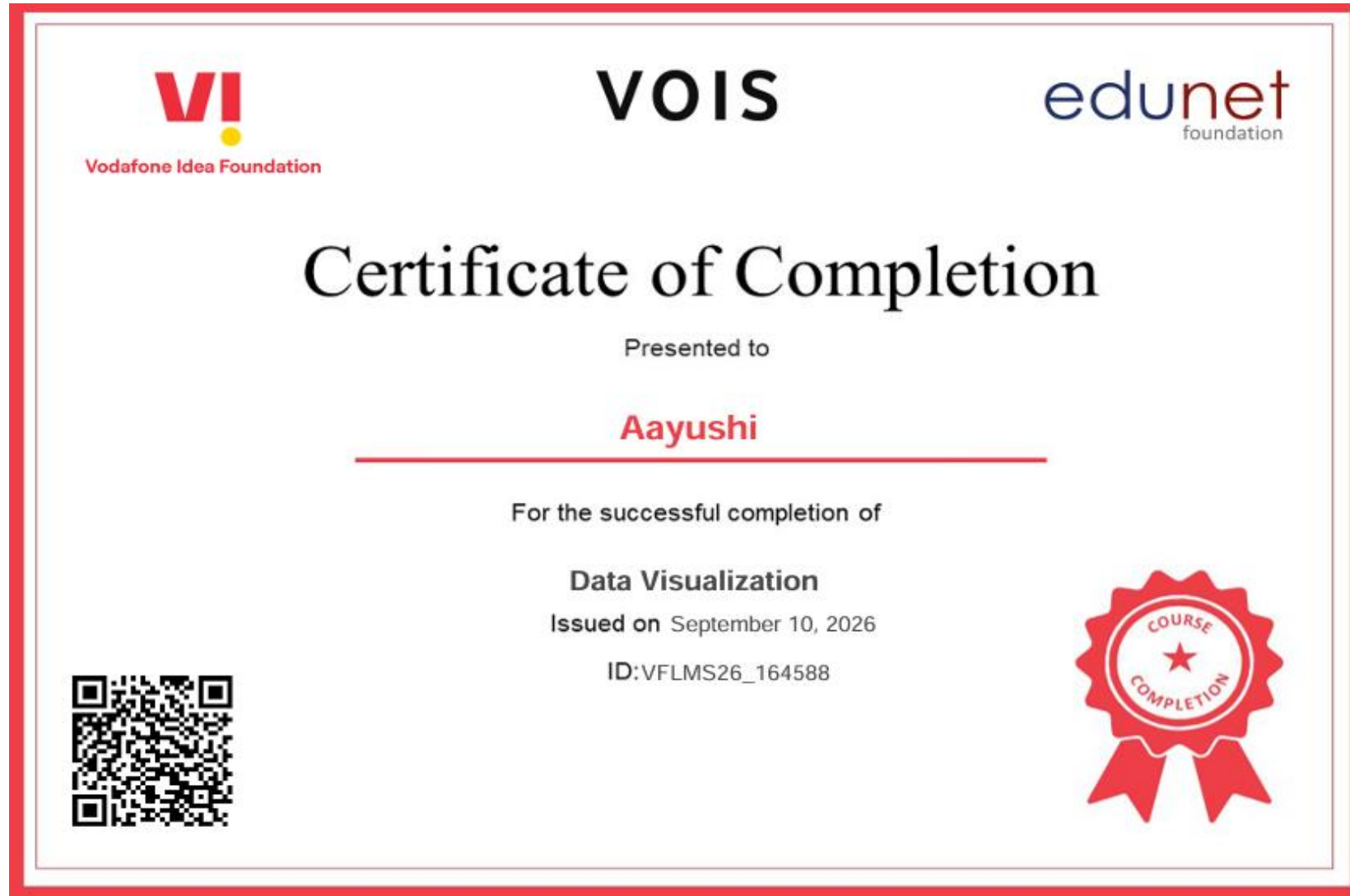


GitHub Link

<https://github.com/aayushiror/VOIS-Major-Project>



VOIS Course completion certificate - Course name- Data Visualization



Thank you